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Method-Independent Detection of Residual Radial Structure in SPARC Dark Matter Halos Beyond Single Smooth Equilibrium Profiles

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ABSTRACT

Single smooth equilibrium halo profiles leave statistically significant, radially organized residuals in roughly one third of SPARC rotation curves. We introduce a minimum-order radial-domain decomposition: the dark-matter contribution is approximated by a minimum-order sum of tiered cored support components (Burkert form), with the order selected by the Bayesian Information Criterion. Across 121 galaxies, a second radial domain is required in ∼35% of cases — a fraction robust to confining component scales to the measured radii (a constraint that simultaneously collapses all three-domain solutions) and stable across three halo families (Burkert 35%, NFW 37%, free-shape Einasto 32%), though, as for any information-criterion threshold, the precise fraction is threshold-dependent. The central result is convergence: galaxies requiring a second domain coincide with galaxies independently flagged by a localized Gaussian-shell decomposition (odds ratio 38; mass-adjusted 37.5), and the domain crossover radius tracks the independently fitted shell radius ($\rho = 0.73$, permutation $p = 0.004$, surviving half-light-radius control). Forced two-domain fits to single-domain galaxies collapse to degenerate configurations, providing an internal null. Where the structure is strong enough to test, the multi-domain selection reproduces on rotation curves re-derived from the original radio cubes by independent software — decisively (ΔBIC = 121 for NGC 2403) and at the same crossover radius — establishing that it is a property of the galaxy, not of the fitting basis. In all 41 two-domain systems the single-profile scale radius lies between the two domain scales (median factor 2.2): single smooth fits are systematically biased summaries, not merely incomplete ones. We interpret the second component as a method- and data-independent marker of a second cored radial mass domain (the term "domain" denoting a radial mass component, with no three-dimensional shape implied, which rotation curves cannot constrain). Of the two components, the diagnostic signal is carried by the *outer*: it remains consistent with single-domain halo scaling, whereas forced decompositions of single-domain galaxies cannot reproduce that behavior. We make no claim about the inner component's relation to the halo scaling locus — a displacement forcing reproduces — and remain agnostic as to physical origin.

Keywords: Dark matter (353) — Galaxy rotation curves (619) — Galaxy dark matter halos (1880) — Astrostatistics (1882)

1. INTRODUCTION

Rotation curves of late-type galaxies are conventionally decomposed into baryonic contributions plus a single smooth equilibrium dark matter halo, described by cuspy profiles motivated by ΛCDM simulations (NFW; **?**), observationally motivated cored profiles (Burkert; **?**), or intermediate forms (Einasto; **?**). The adequacy of a *single* smooth profile is usually assessed only through global fit quality. Localized departures — coherent residual features at particular radii — have been discussed largely in connection with baryonic features ("Renzo's rule"; **?**), but a systematic, model-agnostic census of whether single smooth halos suffice is lacking.

In a companion analysis (Bibb 2026, Zenodo, https://doi.org/10.5281/zenodo.20263016) we modeled localized residual structure in SPARC (**?**) rotation curves with Gaussian mass shells superposed on a Burkert backbone, finding BIC-selected shells in roughly half the sample. That decomposition is used here only as a structurally independent comparison basis, not as a source of physical claims. The present Letter asks a deliberately minimal, complementary question: *what is the minimum number of smooth, concentric, cored ra-*

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dial domains required by each rotation curve? Because any sufficiently flexible component will absorb residuals — so that inferred "structure" may reflect the chosen basis rather than the galaxy — the credibility of either decomposition rests on validation. We therefore test the answer against the structurally unrelated shell decomposition, against mass-driven confounds, against an internal forced-fit null, and — most stringently — against rotation curves of the same galaxies derived independently from the original radio data cubes.

Four properties make the result reportable. First, it is robust: the multi-domain fraction is unchanged when components are forbidden from extrapolating beyond the measured radii, while unstable three-domain solutions are eliminated by the same constraint. Second, it is backbone-independent: NFW and Burkert nulls give the same fraction. Third, it is method-independent: two unrelated decompositions flag the same galaxies and localize structure to consistent radii, and this concordance is not explained by galaxy mass. Fourth, and most stringently, it is *data*-independent where it can be tested: for galaxies whose rotation curves can be rederived from the original radio cubes by independent software, the multi-domain selection reproduces — decisively, and at the same radius. A fit improvement on one curve, however constrained the basis, cannot predict the outcome of an independent derivation from a different instrument; that this prediction succeeds is the strongest available evidence that the structure is a property of the galaxy rather than of the parameterization. We emphasize equally what the analysis does *not* find, and we make no claim about the physical origin or three-dimensional geometry of the detected structure.

2. DATA AND METHODS

2.1. *Sample and baryonic model*

We use the SPARC database (**?**), restricted to galaxies with quality flag $Q \leq 2$ and at least six rotation-curve points after removing points interior to the first reliably measured radius: 123 galaxies meet these criteria, of which 121 yield converged fits. Baryonic contributions adopt fixed mass-to-light ratios $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$ at $3.6\,\mu$m with the gas contribution at unity, combined with signed squares so that $V_{\rm bar}^2 = V_{\rm gas}|V_{\rm gas}| + 0.5\,V_{\rm disk}|V_{\rm disk}| + 0.7\,V_{\rm bul}|V_{\rm bul}|$. Velocity uncertainties are floored at $1\,{\rm km\,s^{-1}}$. Points at which the baryonic model exceeds the observed velocity ($V_{\rm obs}^2 \leq V_{\rm bar}^2$) are excluded from the dark-matter fit, matching the companion analysis.

2.2. *Minimum-order concentric-domain decomposition*

The inferred dark-matter contribution is approximated by a minimum-order sum of cored radial support components of Burkert form,

$$V_{\rm DM}^2(r) = \sum_{i=1}^{N} V_{\rm B}^2(r;\, \rho_{0,i},\, a_i), \tag{1}$$

subject to ordering constraints that enforce a tiered structure and break exchange degeneracies: $\rho_{0,i}$ strictly decreasing and a_i strictly increasing outward. A soft prior ties the total enclosed dark mass at the outermost measured point to the dynamically inferred value. Fits use bounded nonlinear least squares with 20–24 random restarts; model order is selected by BIC $= \chi^2 + 2N \ln n_{\rm pts}$, evaluated for $N = 1$–4. At $N = 1$ the model reduces exactly to a single Burkert halo, providing the null hypothesis within the same machinery.

Two variants are run. In the *unconstrained* variant, scale radii are free. In the *capped* variant, every component is required to satisfy $a_i \in [r_{\rm min}, r_{\rm max}]$, the measured radial range, so that no component's characteristic scale lies where there are no data. The cap is not a technical variant but a *stability discriminator*: structure that survives it is resolved by the data, while structure that collapses is identified as extrapolation artifact. It thereby provides an internal falsification mechanism for each candidate domain.

2.3. *Comparison methods*

Gaussian-shell decomposition. The companion method models localized residuals as Gaussian mass shells of center r_s, width σ, and mass M_s on a single Burkert backbone, with shell count BIC-selected and a width constraint $\sigma/r \leq 0.4$ that defines locality. The preferred fractional width is cap-invariant: the median $\sigma/r \simeq 0.22$ is unchanged as the allowed ceiling is raised from 0.4 to 1.0, with only 4% of shells near the ceiling at a cap of 1.0. The shell and domain decompositions share data but use structurally unrelated basis functions — a localized annular perturbation versus a global cored profile — and are fit independently.

Backbone generalization. The minimum-order analysis of §**??** is repeated with NFW and Einasto in place of Burkert. For Einasto we run two variants: a fixed-shape null ($\alpha = 0.16$) and a free-shape variant in which α is an additional shared parameter ($k = 2N + 1$, with the extra parameter charged to the BIC), bounded to the physical range $\alpha \in [0.1, 0.3]$.